

Vishal Kumar Allam

Chicago, IL | 816-666-2813 | itsmevishal36@gmail.com | **Work Authorization: H1B**

PROFESSIONAL SUMMARY

- Quality Engineering and Test Automation professional with over **6+ years** of experience designing, developing, and maintaining scalable automation solutions across enterprise applications, APIs, microservices, distributed systems, and data-driven platforms.
- Strong hands-on expertise with **Java, TypeScript, JavaScript, Python, C#, Karate, RESTAssured, Playwright, Cypress, Selenium WebDriver, and SQL**, building reliable API, UI, integration, regression, and end-to-end automation frameworks.
- Experienced testing **AI/ML-driven features and data-driven systems**, including functional behavior, test-data validation, edge cases, model-assisted workflows, integration dependencies, and release-readiness scenarios.
- Strong automation-framework engineering experience creating reusable components, utilities, libraries, service clients, fixtures, and test-data abstractions while modernizing legacy manual coverage into maintainable automated solutions.
- Experienced validating **REST APIs, service contracts, microservices, relational databases, distributed workflows, and backend integrations**, with emphasis on reliability, performance, security, data integrity, and error handling.
- Hands-on CI/CD and cloud experience with **Jenkins, GitHub Actions, Azure DevOps, Git, AWS, Docker, Kubernetes, Lambda, S3, CloudWatch, and API Gateway**, supporting continuous quality validation and rapid engineering feedback.
- Analytical and ownership-driven engineer experienced with exploratory testing, root-cause analysis, defect management, release/post-production validation, quality reporting, Agile collaboration, and AI/Dev Assist tools for faster automation development and troubleshooting.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Automation:** Java, TypeScript, JavaScript, Python, C#, SQL, Karate, RESTAssured, Playwright, Cypress, Selenium WebDriver, JUnit, TestNG, NUnit.
- **API & Distributed Systems:** REST APIs, Microservices, Distributed Systems, Contract Testing, Postman, Swagger/OpenAPI, JSON, Backend Validation, Service Workflows, API Integration Testing.
- **Testing Methodologies:** Functional Testing, Integration Testing, Regression Testing, End-to-End Testing, UAT, Exploratory Testing, Edge-Case Testing, Performance Testing, Load Testing, Security Validation.
- **AI/Data Quality:** AI/ML Feature Testing, Data-Driven Systems, Test Data Validation, AI Workflow Validation, Edge-Case Identification, AI-Assisted Testing, Dev Assist Tools, Code Generation, Diagramming, Quality Risk Analysis.
- **CI/CD & Cloud:** Jenkins, GitHub Actions, Azure DevOps, Git, GitHub, AWS, Lambda, S3, CloudWatch, API Gateway, CI/CD Pipelines, Continuous Validation, Release Automation.
- **Data & Infrastructure:** SQL Server, PostgreSQL, MySQL, Relational Databases, SQL Queries, Data Reconciliation, Docker, Kubernetes, Containerized Test Execution, JMeter, Gatling.
- **Engineering Practices:** Automation Framework Design, Reusable Libraries, Root Cause Analysis, Defect Management, Test Reporting, Release Validation, Post-Production Testing, Agile/Scrum, CI/CD Modernization, Continuous Improvement.

Test Automation Engineer

PCMI LLC (Full Time)

Oct 2024 – Present

Chicago, IL, USA

- Analyze complex requirements and technical designs to create comprehensive test strategies covering **functional, integration, regression, exploratory, API, end-to-end, and UAT-focused testing** across enterprise and data-driven applications.
- Design and enhance scalable API and UI automation frameworks using **Java/C#, TypeScript, Playwright, Karate/RESTAssured-style patterns, REST APIs, and SQL**, creating reusable utilities, service clients, fixtures, and validation libraries.
- Validate AI/ML-assisted and data-driven features through structured test data, edge-case scenarios, deterministic business-rule checks, backend verification, and integration testing while documenting quality risks and corrective recommendations.
- Test microservices and REST integrations for service contracts, workflow behavior, authentication, error handling, data integrity, and cross-service consistency using Postman, JSON assertions, SQL queries, and backend diagnostics.
- Integrate automated suites with **Azure DevOps, TeamCity/GitHub-style CI/CD, Git, Docker, and Kubernetes**, enabling continuous execution, parallel regression, build-quality gates, and rapid feedback throughout deployment workflows.
- Investigate automation and application failures using logs, API traces, SQL, test data, and environment evidence while performing root-cause analysis and distinguishing defects from framework, data, or infrastructure issues.
- Support release validation and post-production testing, communicate test results and quality risks, review automation code, and use AI/Dev Assist tools for code generation, troubleshooting, framework improvements, and faster delivery.

Environments: Java, C#, TypeScript, Playwright, Karate/RESTAssured Concepts, REST APIs, Postman, SQL Server, Azure DevOps, Git, Docker, Kubernetes, CI/CD, AI/ML Testing, Data Validation, Root Cause Analysis, Release Validation, Agile/Scrum.

SDET / QA Automation Engineer

Client: CVS

Dec 2023 – Sept 2024

Chicago, IL, USA

- Developed automated quality coverage across healthcare web applications, APIs, microservices, and integrated backend workflows using **Java, TypeScript, Playwright, Cypress, RESTAssured, Selenium WebDriver, and SQL**.
- Designed API automation around REST service contracts, business workflows, negative scenarios, and edge cases using RESTAssured/Postman patterns, JSON validation, authentication checks, and relational database verification.
- Performed functional, integration, regression, exploratory, end-to-end, and user-acceptance readiness testing while validating healthcare business requirements, test data, cross-system behavior, and downstream processing.
- Integrated automated testing with **Jenkins, GitHub/GitLab, Git, and CI/CD pipelines**, supporting continuous validation, fast developer feedback, automated regression, quality gates, and dependable release decisions.
- Investigated test failures using Splunk, application logs, API responses, SQL queries, browser traces, and service diagnostics while collaborating with developers to identify root causes and implement durable fixes.
- Executed performance and load validation using **JMeter/Gatling-style workloads**, analyzing API latency, throughput, concurrency, resource behavior, and service reliability across distributed application components.
- Worked with developers, Product stakeholders, and QA engineers to clarify requirements, identify risks, document defects, report quality status, and apply Dev Assist/AI tooling to accelerate automation development and maintenance.

Environments: Java, TypeScript, Playwright, Cypress, RESTAssured, Selenium WebDriver, Postman, REST/JSON APIs, SQL, Jenkins, GitHub/GitLab, Git, CI/CD, Splunk, JMeter, Microservices, Healthcare Applications, AI-Assisted QA, Agile/Scrum.

Software Engineer

Infor

Sept 2021 – Dec 2022

Hyderabad, India

- Designed and maintained automation frameworks using **Java, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, REST APIs, Python, and SQL** across enterprise web applications and distributed backend services.

- Developed API automation for REST services and microservices, validating service contracts, JSON payloads, business workflows, boundary cases, and backend data using Postman, reusable service utilities, and SQL queries.
- Built reusable automation components, helper libraries, test-data utilities, and framework abstractions that improved maintainability, reduced duplicated test code, and accelerated regression development across multiple teams.
- Executed functional, integration, regression, end-to-end, and exploratory testing while assessing requirement changes, integration dependencies, test-data conditions, and potential quality risks before release.
- Integrated automated suites with **Jenkins, Git, Maven, Docker, and CI/CD pipelines**, providing repeatable regression execution, automated build validation, and rapid feedback throughout Agile software delivery.
- Performed performance and load testing using JMeter and application diagnostics while analyzing response times, throughput, service stability, and failures across enterprise application integrations.
- Investigated defects through logs, API responses, SQL results, and debugging while collaborating with developers on root-cause resolution, modernization of legacy tests, and continuous improvement of automation architecture.

Environments: Java, Python, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, Postman, REST APIs, JSON, SQL, Jenkins, Git, Maven, Docker, CI/CD, Microservices, JMeter, Framework Development, Root Cause Analysis, Agile/Scrum.

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed and executed manual and automated functional, integration, regression, API, and end-to-end testing using **Java, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, REST APIs, and SQL**.
- Created reusable API and UI automation components for enterprise applications, covering positive, negative, boundary, integration, and workflow scenarios while gradually replacing repetitive manual regression with automation.
- Performed REST/SOAP service testing using Postman, JSON/XML payloads, SQL, and backend verification to validate service behavior, data integrity, processing rules, error responses, and integrated application workflows.
- Maintained test cases, test scenarios, test data, execution evidence, and defect documentation while validating application functionality against requirements and supporting regression and UAT activities.
- Integrated automated regression suites with **Jenkins, Git, and CI/CD pipelines**, supporting scheduled testing, continuous build validation, repeatable execution, and earlier identification of software defects.
- Investigated failed test executions through logs, screenshots, API responses, SQL records, and debugging techniques while collaborating with developers to isolate root causes and validate production-ready fixes.
- Worked across concurrent Agile initiatives while participating in requirement discussions, defect triage, release validation, and post-deployment verification and communicating testing progress, risks, and overall quality status.

Environments: Java, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, Postman, REST/SOAP APIs, JSON/XML, SQL, Jenkins, Git, CI/CD, Functional/Integration/Regression Testing, UAT, Defect Tracking, Release Validation, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.